

PROPOSAL

vrCAVE × Meta
VR Game Market Research + Proposal Document v2.1

Tuesday February 6, 2024



Methodology

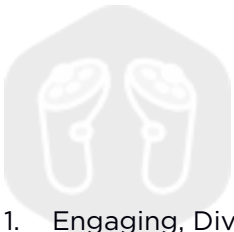
We are interested in developing a game for the Meta Home Store that we feel has the greatest chance of commercial success. We are taking a market-driven approach, rather than strictly a creative-driven approach.

In looking at the game that would be best suited, we considered 3 criteria:

1. **What kind of games does vrCAVE make well?**
We want to play to our strength, and bring our expertise and knowledge of VR to the home market. vrCAVE has successfully self-produced and released 10 VR games for the location-based VR market.
2. **Which games do well commercially, why?**
We collected the number of ratings, considered to be a good indication of a game's performance and applied a formula to estimate revenue based on the price of the game.
3. **Which segment of the market is underserved?**
Focus here will not only increase the chances of our game being a success, but will help in building a stronger VR ecosystem by ensuring that more preferences in games are represented in the store.



Where vrCAVE Excels



1. Engaging, Diverse Gameplay

While most of our content is templated off of the live action Escape Room, each one of our games have multiple activities across multiple unique environments. Each one of these activities lie somewhere on the spectrum from very logic-focused (let's mix the right ingredients to brew this potion) all the way to action-focused (let's work together to take down the menacing fire-breathing dragon).



2. Social/Multiplayer

vrCAVE's mission is to bring people together through technology. To succeed in our games, players must communicate. This creates countless memorable dynamic social situations as groups devise and execute plans and puzzle out solutions together. Through conquering the game, they create their own social story.



3. Diverse Team-Based Mechanics

To emphasize our social focus, we create a diverse set of game mechanics and scenarios that give all players a chance to play a meaningful role, regardless of skill. If someone is not good at aiming, they might help defend, reload, or keep track of targets. Across all of our games, we have amassed experience in a huge variety of different challenges that are fun, easy, and leave no one behind.



4. No VR Sickness

Again given our audience, we place enormous importance on avoiding VR sickness. While we cannot say we have eliminated this in all possible players, we can confidently say fewer than 1 in 200 people have experienced significant discomfort while playing.



5. Taking Full Advantage of VR

When creating experiences we ask ourselves, "what can we do that's only possible in VR?" From full body movements, to fantastic environments, to smart control over what each individual headset shows the players, we seek to use VR at its very best.



6. Reliable Delivery Capability

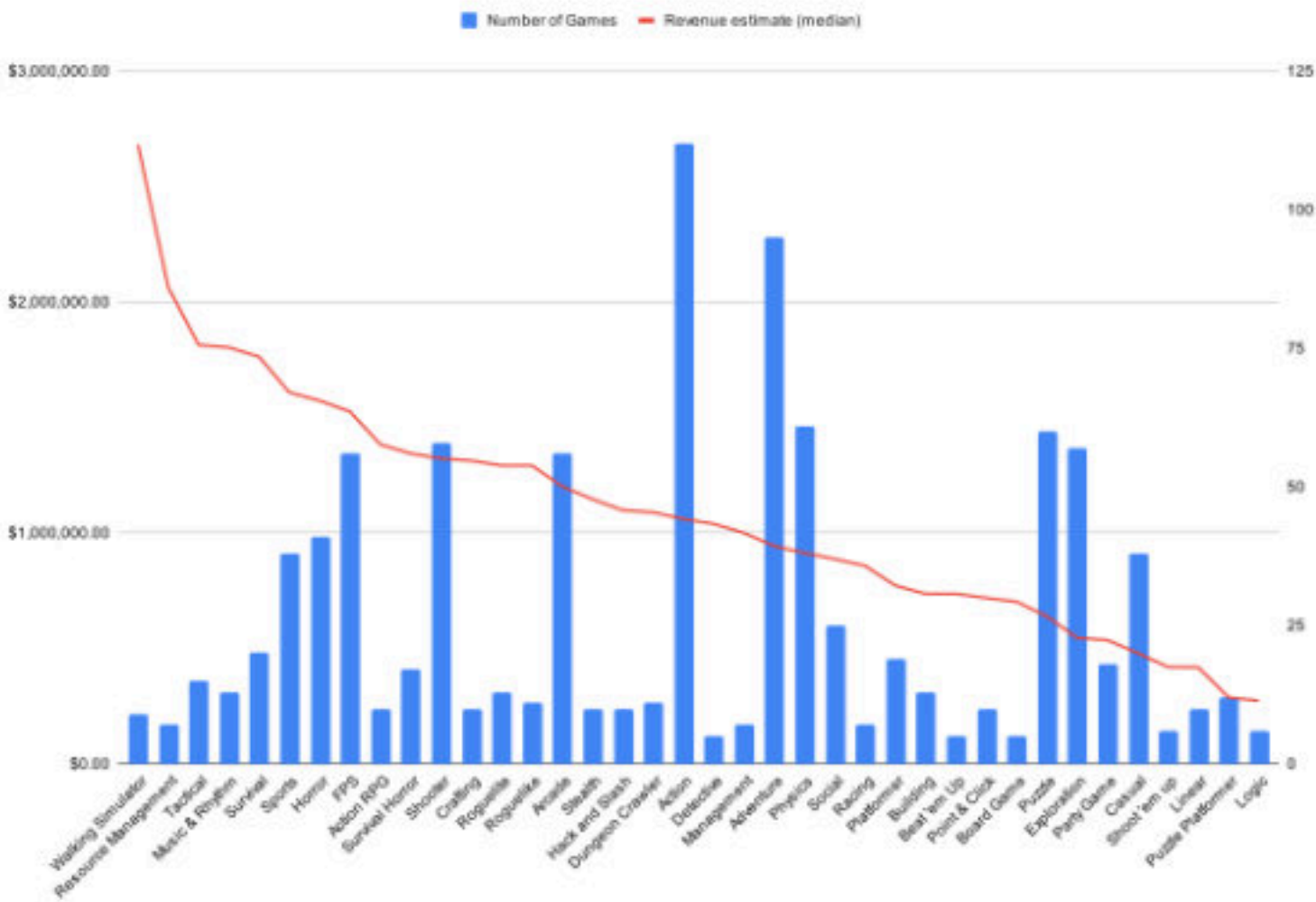
vrCAVE has released at least one new full-length VR escape room per year for its current customers, on top of other improvements and updates, since 2017. Our team is highly experienced and possesses the proven capability of starting from concept and taking a product to release, on time, repeatedly. Our development team and marketing team is entirely in-house and we have localized our games to 5 different languages so far, with more on the way.

Analyzing Commercially Successful Games

We conducted a study of all games released on both the Meta and SteamVR platforms. We combined genre data from both platforms, and added a category of “Party Game” by manually verifying. We collected the number of ratings, considered to be a good indication of a game’s performance and applied a formula to estimate revenue based on the price of the game. We expect the revenue to be somewhat higher for more casual games. We removed any genre with fewer than five titles and produced the following chart with estimates for median revenue.

VR is a new and growing segment of the gaming market. We also looked to the top 10 games of all time on VRChartz to see if there were any obvious VR analogues to them so far: Wii Sports, Super Mario Bros, Mario Kart Wii, Wii Sports Resort, Pokemon Red/Green/Blue, Tetris, New Super Mario Bros, Wii Play, New Super Mario Bros Wii, Duck Hunt.

What unites many of these games, particularly the Wii entries, is extremely intuitive gameplay that is easy to pick up, but has room to build skill and master. We believe these qualities are integral to furthering the mass appeal of VR games, and would point to examples such as Beat Saber as analogues. As Wii Sports provided a gentle introduction to motion controls but provided plenty of depth, Beat Saber allows players to immediately feel like they can do VR — but leaves an abundance of room to improve. We would identify Beat Saber as a genre dominator and would not want to challenge that. Instead, vrCAVE would aim for a genre without a clear leader and build a game that welcomes all, regardless of experience, and gives them enormous scope for mastery.



Identifying an Underserved Market

By plotting the number of games and comparing it to the median revenue in our chart above, we can understand where the market is saturated, where it is not — and when it is not, if there is opportunity.

“Action” as a genre is applied to almost half the games in the list, and as such we can’t draw much of a conclusion about it. Otherwise, we can look at Adventure, Physics, Puzzle, Exploration, FPS, Shooter and Arcade and conclude that there are too many games competing for the available revenue.

Where Does VR Make The Biggest Difference?

A high priority when vrCAVE produces our escape room games is answering the question: “Why is this in VR?”

Good VR games are as much activity-based as they are anything else. A satisfying physical activity distinguishes a good VR game from a non-VR game that uses traditional button controls. In keeping with an intuitive approach, this leads us to reject heavily menu-based games, or games where the gameplay mechanics are too abstracted from the physical activity. Board games, menu-based tactical games, point and click, and the like can therefore be removed from consideration.

This leaves many genres still available. Some of these genres will work better than others for a strong VR game. As a short list: Resource Management, Tactical, Survival, Crafting, Roguelike/lite, Stealth, Hack & Slash, Dungeon Crawler, and Beat ‘em Up. We would add Party Game to the list as we would expect this particular style can pack in several genres into a single game.

Hand Tracking Controls

Hand tracking presents an exciting opportunity for VR games over traditional controllers in traditional games. Support for this feature exists in some games on the Quest platform, but we believe there is much more potential here, especially when focusing on obvious and intuitive gameplay.

A game concept that is compatible with hand tracking is more likely to be a game with a good reason to be VR — and thus a compelling product for VR consumers looking to make the most of their headsets.

Our Primary Game Concept

These considerations lead us to a primary concept: a party game based around a series of intuitive mini games that are easy to pick up, but can grow with the players' experience. Think of the VR version of several popular games: Mario Party, Wii Sports, Kinect Adventures, etc. We strongly believe we are a good fit for this as:

- vrCAVE has ample experience with minigame mechanics throughout our existing games
- The minigame format maximizes opportunity to keep different people engaged
- The party game category revenue potential is very promising
- There is no clear party game category leader
- A party game can be played and supported endlessly, with new minigame additions and different groups of people; additional sequels and DLCs are possible.
- Hand tracking would be an important feature to be added to leverage the most out of the VR headset and give accessibility to the game to the largest breadth of demographics: a game that is intuitive to pick up and play.



Project Moebius: Proposed Game Working Title

Based on our above research, we arrived at a party game consisting of a series of minigames as the best option that fits the conditions of serving an underserved market, falling in line with our proven capabilities and matching a concept our team is passionate about, and presenting VR gameplay that feels great and is easy to pick up, difficult to master.

Brief Overview of Game Concept

This game is a multiplayer mini-game tournament aiming to essentially be the “Mario Party of VR”. Specifically, 1 to 4 players engage in a 30 to 60 minute “tournament” that is a collection of about 6 to 12 randomly selected mini-games (from a total pool of about 40).

Between each mini-game within a tournament, players are transported back into a “stadium” level where points are awarded based on performance in the previous mini-game, player point standings are updated and shown to all, and the next mini-game is randomly chosen via a roulette wheel. Unlike Mario Party, there is no “board game” or board that players move around, roll dice, or anything like that; the focus is instead solely on the mini-games. At the end of the tournament, the player with the most points wins. Mini-games towards the end of the tournament are worth more points (to allow lagging players to catch up) and some mini-games allow you to steal points from other players to keep things interesting.

These mini-games will be created to be immediately obvious, involving fun motion-based mechanics that newcomers can pick up and play. A single player mode would allow a single player to challenge themselves to achieve top scores, unlock additional features, and gain mastery over the mechanics. The “play-and-pass” feature will allow multiple players to play on a single headset if desired.

Target Audience

vrCAVE’s mission is to bring people together through technology. We would target those looking for a social experience in VR, which we see as a growing segment. The game would be accessible to all ages, with the strongest demographic being casual and hardcore gamers 10–40 years old.

Genre/Theme

This party game would focus on fun, frantic gameplay, accessibility, and competition as its core themes.

Unique Selling Points

We do not see an obvious dominant game in this genre in VR yet, and believe that our experience with minigame mechanics as part of our VR escape rooms will present an attractive casual game option that will encourage adoption and spread among friends and families. We intend to make this an “easy buy”.

The game will also feature a ‘Share Mode’ that would offer a single minigame at a time played by each person in the same room in succession — this would allow the game to serve as a party-based introduction to VR to help encourage people to buy their own headsets to play true multiplayer.

New minigames can be added to the base game over time, either as DLC additions or free updates.

Project Moebius Details

The following is provided as an initial concept, and with the caveat that details may change as we conduct further market research, enter full development, and adjust for balance and quality.

Metagame

The metagame is the in-game explanation as to why the players are doing these activities, and serves to tie together what would otherwise be unconnected tasks. We believe a strong metagame concept will help attract people to the game through a focused marketing message, facilitate repeated games, and allow for a structure of unlockables. It will provide stakes, allow for continuity to keep people interested, and be entertaining. At the same time, the metagame will be swift and streamlined during tournaments to keep the focus on actually getting into the minigames.

Our setting will be a televised tournament run by a mysterious and quirky host. The tournaments will take place across a series of floating sky islands around a central colosseum-themed hub island. Islands may be visible from other islands to help everything feel connected. Some islands will be as high as space to allow for zero-g gameplay. Others may be on a vast ocean on the planet's surface. The world doesn't need to be explained — the tournament simply teleports players to the other islands.

Each time a new game is started, the players will have the chance to make some quick decisions to customize their character before beginning, and between each minigame will have an opportunity to pick up additional cosmetic accessories, receive points, and see the rankings. This time may also be used to play bonuses they may have acquired. The host will then select the next minigame, until the tournament ends. The sections between games would be short and snappy to give people a chance to catch their breath, but not overstay their welcome.

- The sky island concept allows each game to be on its own distinct island, which can be themed according to the activity. Some may stick with the Roman/Greek architecture style, others may be ninja-themed, others ocean, etc.
- With the televised theme, players could be given bonus awards for fulfilling extra tasks to impress the audience (ie. in the Tower of Trash outlined below, successfully use a specific item requested by the audience)
- Single player mode would allow players to unlock free play for individual games for practice and advanced challenges
- Other unlockables could include new characters and accessories, new minigames not otherwise immediately available, a trophy cabinet, medals/crowns/other status accessories, and advanced game modes
- Multiplayer mode would have everyone compete at the same time. Pass and Play might bundle several minigames at a time per player so they're actually playing for a couple minutes

Either as the final challenge in a tournament, or as a special unlockable tournament, the players may also be able to face the host in a slightly longer “boss fight” minigame. This can serve as a way for the game to feel “complete”, but the host would ultimately regenerate with a new name to run the next tournament.

Visual Style and Art Direction

This game would aim for a bright, cartoon-like style without coming off as childish. We want the game to be visually fun and approachable for people to feel like there are no barriers to playing it. The central colosseum hub and the majority of minigames would have bright blue skies and saturated colorful characters and items.

Project Moebius Details II

Minigame Spotlight: Tower of Trash

Players are given a time limit of 60 seconds to build a tower out of awkwardly shaped items they are given, high atop a windy mountain.

- Items are random junk like vases, eggs, ramps, boxes, etc.
- Items come along a conveyor belt one at a time, getting increasingly harder to work with
- Sometimes a gust of wind will come and threaten to knock down the tower — the player must brace the tower rather than try to add to it
- As a difficulty mode escalation, the objects could be built on a base floating in a tank of water rather than solid ground. In this mode, items could need to be fished out of the water with the players not knowing what they have until it pops out
- Some objects could be moving, such as motors or living wiggling creatures
- Players could have the opportunity to throw an item to topple another tower rather than use it themselves, balanced by rotating barriers that will sometimes protect players
- At the end of the game, players have to let go of the towers as their hands become collisionless ghost hands, and the towers must successfully settle to count
- The tallest tower wins



Minigame Spotlight: Moving Wall Cutouts

A series of moving walls rush toward the player. Each has a cutout shape the player's hands and head must match in time in order to get through. Between the walls are coins they have to grab for points.

- Walls come in waves of five at a time
- Each wave comes faster and faster
- As a difficulty escalation, the hand and arm positions start moving around as the wall comes toward the player, as do the coins
- As another, walls can start off see-through so the coins are visible, and then become solid.
- As another, multiple coins may be placed, sometimes in inconvenient positions that are more optimal to leave alone
- If the player fails to get the correct position, the wall shatters and they are unable to grab the coins until after the next one
- Between cutout placement and coin placement, the emphasis will be on getting the players to frantically get themselves into ridiculous positions, emphasized by the cutout body shape beyond just the hands and head
- The winning player is the one with the most coins at the end



Project Moebius

40+ Initial Minigame Concepts

Tower of Trash: Players must build towers out of inconveniently shaped objects without them toppling over • *Physics Challenge*

Moving Wall Cutouts: Players have to fit through cutout shapes in walls moving toward them while collecting coins • *Full Body Movement*

Hungry Monster: Players must use their arms to open up the mouth of a monster, catch food, and move rapidly to chew it • *Full Body Movement*

Extreme Whack-a-Mole: Whack-a-mole but players are inside spheres with targets coming out of holes all around them • *Full Body Movement*

Archery Anarchy: Players are given a bow and swarmed with targets all around them • *Shooting*

Safe Crackers: Players listen for the clicks and feel for vibrations to crack into a safe • *Logic & Memory*

Cannons Ahoy: Players have to load, light, aim, and fire cannons at other players’ ships. Could be teams of two • *Full Body Movement*

Slice and Easy: Random items are launched at the player. They must either dodge for 1 point or slice them with a sword for 2 • *Hack and Slash*

Spelling Bee: Bees have invaded the wizard’s tower. The wizards must cast wind spells to drive them out of windows, etc. • *Full Body Movement*

Speed Chef: Players must combine ingredients coming down a conveyor belt into requested recipes for hungry customers • *Logic & Memory*

Point and Shoot: Players are given guns and must shoot moving targets out of the sky • *Shooting*

Code Crackers: In teams of two, one player has to describe a set of procedurally generated shapes to another to crack a code • *Logic & Memory*

Boulder Chase: Players move their arms to run from a boulder racing down a hill • *Full Body Movement*

Quick Draw Duel: Players are given a countdown after which they spin around, draw their gun, and try to shoot their opponent • *Shooting*

Speed Slicer: Players have two seconds per item to make one clean slice each through a specified line. Multiple waves of items • *Hack and Slash*

Ducky Carnival: A rotating pond contains ducks which must be fished for points you only find out once you have them • *Full Body Movement*

Row Boat Race: In teams of two, players row a boat in time with each other to avoid spinning and capsizing • *Full Body Movement*

Overtime Bomb Squad: Players are given a succession of bombs and instructions on which wire to cut with only seconds per bomb before they explode • *Logic & Memory*

Follow The Leader: Players must match the actions presented by a character • *Full Body Movement*

Crazy Cutter: Players have to use swords to cut through weak points on blocks that zoom toward them • *Hack and Slash*

Hot Potato: A potato that will explode must be tossed around the group • *Full Body Movement*

Drummer Time: Players must hit the correct drum in time with the beat • *Rhythm*

Limbo: Players have to bend backwards when limbo sticks come toward them • *Full Body Movement*

Slingshot Shooter: Shooting gallery with slingshot controls • *Shooting*

Slugger Swing: Players have to hit baseballs with bats at the correct angle and speed to have the ball land in specific areas • *Full Body Movement*

Snowball Fight: Scoop snow, roll it into a ball, and throw it at other players while ducking to dodge incoming fire • *Full Body Movement*

Protect the Candle: Players have to keep a candle flame burning from harsh winds blowing from all directions • *Physics Challenge*

3D Marble Maze: A box with a concealed maze inside has to be tilted to allow a marble to make it to the end • *Physics Challenge*

Hang Glider: Players have to navigate their hang glider toward a target • *Physics Challenge*

Dance Moves: Players have to mimic wacky dance moves given by a character • *Full Body Movement*

Speed Sculptor: Players given a ball of clay to morph into a target shape • *Accuracy*

Gesture Artist: A complex 3D path showing things like spirals, animals, faces, etc. is drawn with an orb. Players must mimic the path • *Accuracy*

Ducky Factory: A set of machines produces rubber ducks. Players must keep them running to produce the most ducks • *Logic & Memory*

Pinkies Out: With tentacles instead of hands, players must pour and drink cups of tea • *Physics Challenge*

Fish Fiend: With tentacle hands, players must grab fish from the water around them and feed themselves • *Physics Challenge*

SucShot: Snag collectables with a suction cup crossbow while avoiding hazards • *Shooting*

Yo-yo Dojo: Yo-yo as many times as possible over a bottomless pit • *Physics Challenge*

Shadow Puppeteer: With hand tracking, match a given projected shape • *Accuracy*

Sisyphus Racers: Race to get a massive, heavy boulder up the side of a mountain while avoiding other players • *Full Body Movement*

Spaceship Mechanic: Players must diagnose and repair a faulty spaceship • *Logic and Memory*

Ricochet Rivals: Dodge-ball style game where players have to avoid or defend themselves from ricocheting projectiles from the environment and each other • *Full Body Movement*

Laser Gazer: Players use their laser eyes to cut out a series of increasingly difficult object shapes. Whoever finishes the most shapes wins • *Accuracy*

Ice Cream Vendor: It’s raining ice cream. Catch the right order of ice cream on your cup or cone to satisfy your customer. Most customer orders fulfilled wins • *Full Body Movement*

Balloon Fill Roulette: Players take turns pumping air into a balloon. Whichever player makes the balloon explode loses • *Luck*

Dino Egg Relay: Players must work together to pass a dinosaur egg to each other using only giant spoons • *Physics Challenge*

Rock, Planet, Potato: Rock, Paper, Scissors, but every option looks identical. The Host decides which move each player has made • *Luck*

Project Moebius Details III

Technical Specifications

We would target the Meta Quest 3 as the primary platform and optimize our build for this headset. We will also look to support Meta Quest 2 given its current popularity.

Development Time

December 2023 to February 2024: Market research, design, and hiring

June 2024: Public demo

Late 2024: Public beta release

Winter/Spring 2025: Official Launch

Budget

Expected Development Cost: \$620,000 CAD

Expected Marketing Cost: \$151,000 CAD

Marketing & Promotion

vrCAVE will begin marketing and promoting this game when the public demo is announced. We will announce at a major local gaming event: Game Con Canada. Entering its second year in 2024, the event expects 40,000 - 60,000 attendees and will serve as an excellent opportunity for us to kickstart our promotion, which will be focused around community building and wishlist acquisition. We will position this game as the answer for those seeking a more social VR experience.

We are researching how other games in the social multiplayer space have been successful in their promotion. Our plan will involve:

- Influencer outreach
- TikTok, Instagram, Facebook, Discord marketing and community building
- Paid advertisement
- Regular updates to store pages pre- and post-release
- Email list building and engagement

Risk Assessment

- In our analysis, there were a fair number of casual multiplayer games with modes that match the idea of a “party game”. We note that the closest match to our concept, Loco Dojo, does not appear to be a major hit according to the revenue estimate formula, however:
 - At 6 years old, Loco Dojo was released to a much different VR landscape
 - The formula likely underestimates revenue for more casual games
 - There does not appear to be a major winner in this category yet
- Multiplayer games require other players to be available
 - This game's multiplayer will be playable with as few as two players
 - Single player mode will give players another option
- For in-person play, few people own multiple headsets
 - Share Mode will allow an in-person gathering to compete by taking turns, or free play in single player mode

THE HOST

Expressive
Nostalgic
+Sassy

A comedic announcer housed in a floating CRT TV that keeps the action frantic, and the players in the game.



The Host

The Host displays a wide range of emotions to heighten the frantic party action. A bit bumbling (and perhaps malicious, it's hard to tell) he looks to make great entertainment no matter the cost.



THE COLOSSEUM

The hub where players compare scores and take breather in-between minigames. Players are transported from it to minigame islands in the nearby sky.



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